

A Comparative Study of Downscaled Canopy Reflectance

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Abstract

We examine the behaviour of methane fluxes in alpine catchments. Calibration drift over the trial period remained below the threshold set out in the protocol. Sensitivity to the smoothing window proved modest across the range we examined. Residual error grows approximately linearly with the number of coupled parameters.

1. Introduction

The measurements reported here were collected over twelve independent trials under controlled conditions. Replicate agreement was high, suggesting the protocol is robust to small handling differences. We release the processing scripts and intermediate artefacts to support independent verification. Residual error grows approximately linearly with the number of coupled parameters. Calibration drift over the trial period remained below the threshold set out in the protocol. A modest but repeatable bias appears in the lowest-magnitude bin and warrants further study. The reduced model retains most of the explanatory power at a fraction of the computational cost. Our baseline reproduces the behaviour described in earlier work to within the stated tolerance.

Residual error grows approximately linearly with the number of coupled parameters. Longer observation windows narrow the confidence interval but do not shift the central estimate. Replicate agreement was high, suggesting the protocol is robust to small handling differences. Our baseline reproduces the behaviour described in earlier work to within the stated tolerance. The measurements reported here were collected over twelve independent trials under controlled conditions.

We release the processing scripts and intermediate artefacts to support independent verification. Taken together, these findings argue for treating the two regimes separately in future analyses. A modest but repeatable bias appears in the lowest-magnitude bin and warrants further study. Sensitivity to the smoothing window proved modest across the range we examined. Our baseline reproduces the behaviour described in earlier work to within the stated tolerance. The measurements reported here were collected over twelve independent trials under controlled conditions. Replicate agreement was high, suggesting the protocol is robust to small handling differences. We observe a pronounced inflection once the load exceeds roughly sixty percent of nominal capacity.

2. Related Work

Discrepancies at the boundary are consistent with the simplifying assumptions noted above. The measurements reported here were collected over twelve independent trials under controlled conditions. Neither of the alternative formulations improved on the simple additive model in our setting. Longer observation windows narrow the confidence interval but do not shift the central estimate. The reduced model retains most of the explanatory power at a fraction of the computational cost.

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3. Materials and Methods

Sensitivity to the smoothing window proved modest across the range we examined. Discrepancies at the boundary are consistent with the simplifying assumptions noted above. Longer observation windows narrow the confidence interval but do not shift the central estimate. Replicate agreement was high, suggesting the protocol is robust to small handling differences. Taken together, these findings argue for treating the two regimes separately in future analyses. Our baseline reproduces the behaviour described in earlier work to within the stated tolerance. Residual error grows approximately linearly with the number of coupled parameters. We release the processing scripts and intermediate artefacts to support independent verification.

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4. Experimental Setup

Taken together, these findings argue for treating the two regimes separately in future analyses. The measurements reported here were collected over twelve independent trials under controlled conditions. Neither of the alternative formulations improved on the simple additive model in our setting. A modest but repeatable bias appears in the lowest-magnitude bin and warrants further study. Our baseline reproduces the behaviour described in earlier work to within the stated tolerance. Residual error grows approximately linearly with the number of coupled parameters. We observe a pronounced inflection once the load exceeds roughly sixty percent of nominal capacity. Sensitivity to the smoothing window proved modest across the range we examined. Discrepancies at the boundary are consistent with the simplifying assumptions noted above.

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5. Results

A modest but repeatable bias appears in the lowest-magnitude bin and warrants further study. The measurements reported here were collected over twelve independent trials under controlled conditions. The reduced model retains most of the explanatory power at a fraction of the computational cost. Calibration drift over the trial period remained below the threshold set out in the protocol. Taken together, these findings argue for treating the two regimes separately in future analyses.

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6. Discussion

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